

UCab — Full-Stack MERN Cab Booking Platform

Solo Developer / Team Member: Shaik Sumiya Zainab | Project ID: N/A (Solo Track)

Phase 4: Project Planning Phase — Comprehensive Project Planning & Timeline

Project Name: Cab Booking (`UCab`)
Project ID: `N/A (Solo Track Submission)`
Team Member / Solo Developer: Shaik Sumiya Zainab

1. Solo Developer Sprint Schedule

Because the entire project was built individually, agile sprint planning was adapted into structured sequential development phases over a 2-week intensive build cycle.

```
gantt
  title Solo Full-Stack Development Timeline (UCab)
  dateFormat YYYY-MM-DD
  section Phase 1: Foundation & Setup
    UX Research & Empathy Mapping      :done, p1, 2026-07-01, 2d
    Express API Setup & Mongoose Schemas :done, p2, 2026-07-03, 3d
  section Phase 2: Core Features
    Tabbed Auth & Bcrypt Security Layer :done, p3, 2026-07-06, 2d
    Fleet Catalog & Distance Fare Engine :done, p4, 2026-07-08, 3d
  section Phase 3: Advanced Perks & Admin
    In-Ride Refreshments & Charity Add-ons :done, p5, 2026-07-11, 2d
    Admin KPI Dashboard & Fleet CRUD      :done, p6, 2026-07-13, 2d
  section Phase 4: Resilience & QA
    Zero-Config Fallback Storage Engine   :done, p7, 2026-07-14, 1d
    Documentation & GitHub Repository Sync :done, p8, 2026-07-14, 1d
```

2. Effort Estimation & Resource Allocation Table

Task ID	Component / Deliverable	Estimated Hours	Actual Hours Spent	Primary Complexity / Challenge
TSK-01	UI/UX Wireframing & Glassmorphism Design System (`index.css`)	12 hrs	10 hrs	Achieving responsive grid balance across mobile and desktop.
TSK-02	Express.js REST API Setup & Mongoose Schema Modeling	15 hrs	14 hrs	Structuring clean relational references between User, Car, and Booking models.
TSK-03	JWT Bearer Token Security & Role-Based Middleware	10 hrs	10 hrs	Ensuring separate token validation paths for standard Riders and Super Admins.
TSK-04	Interactive Booking Checkout & Deterministic Distance Calculator	16 hrs	18 hrs	Integrating dynamic promo discounts (`UCAB20`) and add-on perk checkboxes.
TSK-05	Live GPS Progress Bar & Trip Status Synchronization	14 hrs	12 hrs	Mapping exact state transitions from admin dispatch to user tracking screens.
TSK-06	Executive Admin Dashboard & Multi-Image Fleet CRUD	18 hrs	16 hrs	Building clean `AddCar.jsx` and `EditCar.jsx` forms with availability toggles.
TSK-07	Zero-Config Dual-Engine Fallback Adapter (`store.js`)	16 hrs	16 hrs	Handling atomic file synchronization (`data.json`) without schema validation errors.
TSK-08	Exhaustive 8-Phase Documentation & Template Compliance	12 hrs	12 hrs	Producing comprehensive documentation reports across all 8 required project folders.
TOTAL	**Full-Stack Solo Delivery**	**113 hrs**	**108 hrs**	**Successfully Delivered On Time ' **

3. Risk Management & Mitigation Strategies

Risk Description	Severity	Probability	Solo Developer	Mitigation Strategy
Local MongoDB Unavailability During Viva Presentation: Database daemon crashes or port 27017 is blocked on the grader's PC.	**Critical**	**High**	**Zero-Config Storage Engine:** Built `server/db/store.js` which detects connection errors within 2 seconds and automatically writes to `data.json`, ensuring 100% demo success.	
Scope Creep on Third-Party Payment Gateway Integration: Spending excessive time configuring Stripe API secret keys.	**Medium**	**Medium**	**Simulated Automated Checkout:** Implemented a clean, frictionless saved payment profile selector that verifies checkout state locally without external API latency.	
State Desynchronization Between Admin Dispatch & User Tracker: User screen doesn't update				

after admin changes trip state. ****High** **Low** **Centralized API Queries:**** Designed
`UserHome.jsx` and `MyBookings.jsx` to fetch active booking states dynamically on mount and tab
switch.